

Representation of Logic Gates Using Quiver Theory

Academic Project
submitted to the

Shaheed Rajguru College of Applied Sciences for Women,
University of Delhi
for the partial fulfilment of the degree of

B.Sc.(Hons.) Mathematics with Research

By

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May, 2026

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Abstract

Logic gates represent the basic components of all digital systems, and thus, modern computing is based upon their existence. They have numerous applications, especially in computer architecture, digital electronics, and artificial intelligence. The standard technique of representing logic gates is Boolean algebra, which is accurate in terms of logic but does not provide sufficient mathematical structure for internal analysis.

Another method used to model logic gates is neural networks, which provides flexibility but relies on approximations and requires training, which cannot ensure mathematically precise logical outputs. Therefore, they are not suitable for applications where complete accuracy is required.

In this project, we propose a mathematical analysis of logic gates through the use of quiver representation theory. In a quiver, the vertices are associated with real vector spaces, while the arrows are assigned linear transformations with scalar values. This model enables the representation of logical operations mathematically without the need for approximation or training techniques.

Representations are constructed based on quivers for each of the seven basic binary logic gates AND, OR, NOT, NAND, NOR, XOR, and XNOR. For the single-layer logic gates, a quiver representation with only one layer is constructed, while the non-linearly separable logic gates XOR and XNOR are represented by a two-layer quiver model. A novel concept of sensitivity analysis is proposed to establish the full range of the bias values within which a logic gate generates correct logical output for all possible combinations of binary inputs. It is shown that a non-empty bias interval exists for each logic gate, and its width equals exactly 1.

Outputs generated by each quiver model for all combinations of binary inputs are checked against the corresponding Boolean truth tables, showing full logical equivalence with the original logic gates. The analysis shows that despite their different logical behaviors, the quiver representations of all single-layer logic gates share the same underlying directed graph structure, and the only difference between them lies in their weights and bias intervals.

Keywords: neural network, quiver representation, logic gates, boolean algebra

Contents

1	Aim	1
2	Review of Literature	1
2.1	Foundations of Boolean Logic and Digital Circuit Design	1
2.2	Neural Networks and the Modeling of Logical Operations	2
2.3	Quiver Representation Theory	4
2.4	Connection Between Neural Networks and Quiver Representations	5
3	Objective	6
4	Research Methodology	7
4.1	Overview	7
4.2	Mathematical Preliminaries	7
4.3	Neural Network Analysis	9
4.4	Quiver Framework Development	10
4.5	Sensitivity Analysis	12
4.6	Validation	14
5	Results & Discussion	14
5.1	Single-Layer Gate Representations	14
5.2	Two-Layer Gate Representations	22
5.3	Summary of Results	28
6	Conclusion	29
7	Further scope of Research	30

1 Aim

The main purpose of the study is to create an appropriate algebraic background for the description of logic gates using the representation theory of quivers. The idea is to view logical entities as vector spaces and relations between them as linear transformations, which would give us a consistent point of view on how logical results can be obtained from algebraically constructed systems. The study aims at transcending symbolic Boolean representations and creating a representation-theoretic model of computations.

2 Review of Literature

In the current study, we rely on a wide variety of sources in areas such as representation theory, neural network theory, Boolean logic, circuitry, and algebraic computations. The discussion follows the logical sequence of the current investigation, starting from the classical roots of logic gate and digital circuits, through the advantages and disadvantages of using neural networks for implementing logical operations, to the mathematics behind quiver representations, and, ultimately, to the link between neural networks and quiver representations, which forms the basis of the current study.

2.1 Foundations of Boolean Logic and Digital Circuit Design

Classical theory of logic gates is developed from the algebraic formulation of logical operations by George Boole [8] during his pioneering study on laws of thought. This formulation of logical operations in terms of Boolean algebra has continued to be the standard notation used for describing logical behavior of electronic logic gates. The theory is further developed and generalized by Shannon [46], who proved that the Boolean algebra could be used to analyze and design relay and switching circuits and hence provided a basis for electronic design of digital systems, thus initiating the discipline of digital circuit theory.

The concept of multi-valued logic was suggested by Rescher [41]. Multi-valued logic is considered a generalization of classical binary logic since it allows logical variables to have more than two states (values). The paper is useful in terms of the extension of the current approach beyond classical binary logic. In this context, Akers [1] presented the use of binary decision diagrams as a compact and canonical form of Boolean functions, while Bryant [9] provided graph-based methods for Boolean function manipulations. Both references serve as an example of the importance of using algebraic and combinatorial structures for representation and analysis of logical functions, providing an excellent background for developing a new approach based on quivers.

Classical texts that treat digital logic design theory include books by Hayes [18].

The mathematical approach to logic, treating it as a vector space, has been initiated by Westphal and Hardy [50], where logical propositions were introduced in vector spaces and logical connectives were considered as linear or affine operators. The current work is inspired by this publication, which serves as a conceptual link between the classical approach to logic and the linear-algebraic approach implemented here.

The book written by Harrison [17] contains a detailed discussion on practical logic and automation of reasoning about logical systems. Further classical treatments of digital logic design theory include books by Kohavi and Jha [24], Rafiquzzaman [38], and Mano and Ciletti [28]. These books offer a comprehensive description of combinational and sequential logic, truth tables, logic minimization, and implementation of logic gates in circuits, while also providing an introductory treatment of digital logic and microcontrollers. Together, these books provide the classical theory of digital logic design to which this project contributes.

2.2 Neural Networks and the Modeling of Logical Operations

The mathematical basis of neural network models of logical computations can be traced back to the initial work of McCulloch and Pitts [29]. They introduced one of the earliest mathematical models of a neuron, which operates as a unit for logical calculations by performing weighted sums of inputs and applying a step function to generate binary output values. Such a description resembles the single-layer quiver design presented in the current paper and defines the historical roots of the connection between logical computation and weighted linear mappings over vector spaces.

Back propagation algorithm, the default method for training multi-layer neural networks, was developed by Rumelhart, Hinton, and Williams [43]. This landmark study paved the way to show that training deep neural networks is computationally feasible, and forms the foundation of most contemporary methods of neural network training. Hinton [22] examined the representation learning in multiple layers, suggesting that the hierarchies of abstractions extracted from the inputs form a characteristic feature of the processes carried out by deep neural networks. This approach has been used in the design of quiver structures involving multiple layers developed in the current project to model XOR and XNOR gates.

The Representation Learning view has been championed by Bengio, Courville, and Vincent [7], with their detailed survey on representation learning showing that although such representations can be easily adapted, they are statistically generated and not algebraically generated and do not ensure logical correctness. In tandem with these architectural insights, Srivastava et al. [47] suggested applying dropout to prevent overfitting.

The problem of optimizing neural networks was further considered by Neyshabur et

al. [34], who proposed path-normalized optimization – an approach to training that is invariant to the scaling of weights. Around the same time, Ioffe and Szegedy [23] introduced batch normalization for training acceleration purposes, and the review of deep learning and its applications by LeCun, Bengio, and Hinton [25] gave a comprehensive overview of the field.

Further architectural and optimization breakthroughs followed as He et al. [19] introduced deep residual learning for image recognition and showed the strength of very deep network architectures. The contemporary theory of neural networks was also covered in a textbook by Goodfellow et al. [16], which provides extensive details about neural network structures, activation functions, learning techniques, and the ability of neural networks to approximate logical functions. The book served as the main source of information while concluding the neural network analysis in Phase 2 of the proposed methodology. It is especially important as it highlights the critical shortcoming of using neural networks for logical computations - their approximative nature and the necessity of training rather than exact specification of logical operations using mathematical equations.

Subsequently, Dinh et al. [12] looked at how deep networks generalize sharp minima, while Sabour et al. [44] introduced dynamic routing between capsules. These examples show the progress in the area of neural network architecture research, while at the same time, they emphasize the need to find another way of building the model that will bring exactness and understanding into the process. Mitchell [32] additionally offers a thorough introduction to machine learning during this period.

Li et al. [26] focused on visualization of the loss function landscape. The aforementioned works related to optimization make evident the mathematical complexity and lack of visibility in the training procedure, which highlights the importance of using the quiver framework because it does not require any training but only specifies weights and biases algebraically.

Following this, Patil and Deshmukh [36] discussed the use of neural networks in implementing logic gates based on the concept of a perceptron. In particular, their study shows that one-layer perceptrons are capable of implementing linearly separable logic functions (such as AND, OR, NOT, NAND, and NOR), while two-layer perceptrons should be used to implement non-linear functions (such as XOR and XNOR). This research paper is a direct predecessor of the neural network analysis performed in the second phase of the current methodology. At the same time, Meng et al.[31] proposed G-SGD optimization algorithm for positively scale-invariant space of ReLU networks. Finally, Raghu and Schmidt [39] have offered an overview of deep learning applications for scientific discovery.

2.3 Quiver Representation Theory

The papers of Peña and Gabriel [37] on wild and tame algebras are pertinent to the classification of logic gate systems based on their representation types. Rotman [42] gives a rigorous introduction to group theory, which is essential for studying quiver representations. More recent developments in the direction of quiver theory during this period include the works of Nakajima [33] concerning varieties attached to quivers and quantum groups, and Hille [21] concerning toric quiver varieties.

The main linear algebraic sources for the current research were selected among the most popular linear algebra books written by Friedberg, Insel, and Spence [14]. The mathematical theory behind the representations of quivers is elaborated extensively in the classic reference by Assem, Simson, and Skowroński [4]. This reference book offers a rigorous and thorough exposition of representation theory of associative algebras through quivers and relations. The present work adopts definitions from this book for quivers, representations, morphisms of representations, indecomposable representations, and methods based on homology. Also used from this book are the notions of path algebra of quiver, bound quiver with relations, and classification of representations based on their indecomposable decompositions.

Following this, Reineke [40] focused on framed quiver moduli and cohomology. Another valuable resource about quiver representations is provided by Schiffler [45], offering a more accessible and intuitive approach to the subject. This source served well in giving insight into the connection between directed graph structure and linear algebraic data, and thus was helpful in grasping the idea of the flow of information through a quiver. Morphisms of representations and equivalence of representations are discussed in this source, providing background for the discussion of circuit equivalence in the present paper.

For an easy and up-to-date overview of the representation theory of algebras, Barot [6] is recommended. The text provides a brief survey of links between quiver representations and different branches of algebra. Barot's textbook added to the comprehensive exposition of Assem et al. and helped to comprehend the modern representation theory of quivers. Further foundational texts of this period include Andrilli and Hecker [2] on elementary linear algebra, and Domokos and Joó [13] concerning toric quiver varieties. Gallian [15] is a fundamental book on abstract algebra that provides a broad view of the theory of groups, rings, and fields and their practical applications.

Das, Manikandan, and Raghavendra [11] later contributed regarding the holomorphicity of moduli of quiver representations. Chindris and Kline [10] investigated semi-stability of quiver representations. Finally, Li [27]'s recent contributions to the topic of quiver algebras and their representations are particularly important since they deal with general

quivers.

2.4 Connection Between Neural Networks and Quiver Representations

The relationship between neural networks and quiver representation theory forms the theoretical underpinning of this current project. Historically, Wood and Shawe-Taylor [51] investigated the relation between representation theory and invariant neural networks, showing the connection between group representation theory and neural network symmetry. Thus, it is shown how algebraic representation theory can serve as an appropriate tool for studying neural networks.

Following this, the work by Nitta [35] introduced a complex-valued back-propagation algorithm, showing that neural networks can be considered over fields different from the field of reals. The latter has connections with quantum extensions of quivers that we will discuss; in particular, vector spaces can be considered not over the real field but over the field of complex numbers.

The paper by Stagge and Igel [48] next focused on neural network structures and their isomorphisms, exploring the space of weights of neural networks in terms of structural equivalence. Their study on isomorphisms of neural networks is directly related to the discussion of quiver morphisms and circuit equivalence in our study, showing that structural equivalence—an important notion in quiver representation theory—has its natural interpretation in the case of neural networks. Healy and Caudell [20] subsequently created a mathematical semantic model of neural networks using category theory. It gives an example of how category theory can be used for describing neural networks, just as quiver theory does for this project. This application of category theory to neural networks precedes the discussion of categorical extensions of quivers in the further scope of this project.

Symmetries in deep networks were later studied by Badrinarayanan, Mishra, and Cipolla [5]. Isomorphic neural networks have also been proposed by Meng and Zhang [30] for graph representation learning. This study highlights the applications of algebraic concepts such as isomorphism to neural network designs, establishing the relation between the representation-theoretic concept of isomorphism and graph learning. As mentioned above, representation theory studies quivers, and the concept of isomorphism is crucial in it.

The foundational bond linking these domains was then rigorously established by Armenta and Jodoin [3]. In their paper, it is shown that a neural network can be modeled exactly as a quiver representation, whereby neurons and layers form vector spaces, weights

are modeled as linear transformations from one vector space to another, and the process of feeding the inputs through the neural network layer by layer can be modeled exactly as composing these linear transformations according to the structure of the underlying quiver graph. This work is the most pertinent of all primary sources for the current research project. It shows how the structure of a neural network model of a logic gate can be represented as algebraic structure of a quiver representation. Furthermore, it shows how the quiver representations can be analyzed using the representation theory methods. This work continues that idea, using the established relation between a neural network and an algebraic structure specifically for the purposes of logic gate modeling. Exact quiver representations of all seven basic binary logic gates, as well as their sensitivity analysis, which allows one to establish the range of possible parameters for each of them, were constructed in the course of research.

One of the more straightforward uses of quiver representation theory in recent studies relating to neural network modeling is the research conducted by Wanditra et al. [49], on artificial neural networks modeled by finite cyclic groups. The work highlights the usefulness of quiver representation in the context of organizing and analyzing different neural network architectures and further establishes that the use of quiver representation theory in understanding neural networks is an emerging field within mathematics. The significance of this work lies in the fact that, while the use of quiver representation in studying neural networks is a well-known practice, the ability to design them using quivers is novel.

3 Objective

The main goal of this research work is to construct a deterministic mathematical system for logical computations. This study consists of the following main objectives:

- Conduct an extensive analysis of quiver representation theory to understand how directed graphs, real vector spaces, and linear transformations form the foundation of computational logic models.
- Explore mathematically the link between the layers of a neural network and quiver objects whereby neurons will be represented as vector space vertices while weights are mapped as linear transformations.
- Develop exact algebraic equations for linearly separable gates, namely: AND, OR, NOT, NAND, and NOR, with specified biases and parameters that produce accurate binary output with 100 % accuracy.

- Apply the quiver method to develop multi-layer logic gate equations which model non-linearly separable gates, e.g., XOR and XNOR through the use of hidden activation functions and intermediate linear maps.
- Define the notion of "Sensitivity" in quivers and find out what the exact "Bias Intervals" (or "Quiver Conditions") are that form the boundaries of the logical decision making process.
- Evaluate the strengths and weaknesses of the method proposed here, particularly considering its potential for scalability towards combinational logic circuits like adders and multiplexers.

4 Research Methodology

4.1 Overview

This work uses theoretical and analytical research methodology. Since the core of the investigation is centered on mathematical modeling and algebraic representation, experimental implementation and practical testing are irrelevant to the chosen methodology. All the stages of the investigation are based on ensuring logical succession, conceptual clarity, and mathematical rigidity. This methodology comprises two significant parts. First, it involves the creation of quiver representations of logic gates, wherein the functioning of each of the gates will be described by precise mathematical terms such as directed graphs, vector spaces, and linear maps. Second, sensitivity analysis will be performed to explore the impact of bias parameter on the logic of each gate's functioning and reveal all the biases necessary for proper logic operation. Both these processes will be executed in four stages: building a basis, analysis of neural networks, creating a quiver framework, and performing sensitivity analysis with verification.

4.2 Mathematical Preliminaries

Before we introduce the framework, it is important to first define the essential mathematical constructs on which the entire methodology is based. This section introduces the terminology used in the project.

Quiver

A quiver Q is a directed graph consisting of a tuple $Q = (Q_0, Q_1, s, t)$, where Q_0 is a finite set of vertices, Q_1 is a finite set of arrows, and $s, t : Q_1 \rightarrow Q_0$ are maps specifying the source and target vertices of an arrow, respectively (Armenta and Jodoin[3]). Quivers

differ from regular graphs by allowing arrows pointing from one vertex to another and loops of arrows on the same vertex. Vertices in this particular research denote processing units such as input neurons, hidden neurons, and output neurons, whereas arrows symbolize the directed transmission of information between these units.

Representation of a Quiver

The representation of the quiver Q over the field of real numbers takes the real vector space $X(v)$ for each vertex $v \in Q_0$ and a linear transformation $X(\alpha) : X(s(\alpha)) \rightarrow X(t(\alpha))$ for each arrow $\alpha \in Q_1$, where $s(\alpha)$ and $t(\alpha)$ denote source and target vertices respectively. The linear transformation transfers and transforms information as it moves from the source to the target vertex through an arrow.

In our project, every vertex gets a one-dimensional vector space represented by the real number line $\mathbb{R}$. Therefore, the linear transformation corresponding to an arrow becomes the multiplication by a real number, or simply a scalar multiplication. The scalar can be identified precisely as the weight assigned to the edge.

Activation Output Formula

In this work, the implementation of the logic gates takes place via quiver-based neural network theory, and the computational structure involves the use of directed graphs and algebraic operations. The key feature of this approach is the utilization of the quiver Q , which refers to a directed graph where every vertex v is assigned a particular vector space W_v and the arrows α define the linear maps $W_\alpha : W_{s(\alpha)} \rightarrow W_{t(\alpha)}$. The mentioned mappings act as deterministic weights that transfer signals from vertex $s(\alpha)$ to vertex $t(\alpha)$. The state of the network is provided by means of activation output $a(W, f)_v(x)$, which can be described as the signal on vertex v under the condition of the input vector $x \in \mathbb{R}^n$. The calculation of this quantity requires the summation of the effects of all received signals, which form the set ζ_v (all arrows such that $t(\alpha) = v$):

$$a(W, f)_v(x) = f_v \left(\sum_{\alpha \in \zeta_v} W_\alpha(a(W, f)_{s(\alpha)}(x)) \right) \quad (1)$$

The nonlinear activation function f_v is then applied to the sum for transformation into a binary signal. Thus, the couple (W, f) consisting of all linear mappings and vertices becomes a solid algebraic basis for the operation of logic gates.

Step Activation Function

The activation function being used in each vertex within this project is the step function and it can be mathematically described as:

$$f(z) = \begin{cases} 1, & \text{if } z \geq 0 \\ 0, & \text{if } z < 0 \end{cases} \quad (2)$$

The reason for selecting this particular activation function is that the function generates deterministic results where the output is always in the form of a binary number (0 or 1) regardless of the input value. The output generated by the step function does not involve any sort of approximation unlike the sigmoid and ReLU functions where an approximated output value is generated based on the input values. Thus, this function suits our model in order to simulate binary logic gates using artificial neural networks.

$$z = \sum_{i=1}^n w_i x_i - b$$

4.3 Neural Network Analysis

In this section, the use of neural networks in the modeling of logical operations, considering their structure and functioning is analyzed thoroughly. Such an examination was essential to establish the essence that needs to be encoded into the quiver framework, as well as determine the exact source of limitations inherent in the neural network approach. Single layer neural networks or single-layer perceptrons, are able to represent logic gates, the function of which is linearly separable. Linearity in this case implies that there exists one linear border, which separates the space of inputs giving outputs of 1 from the space of inputs generating outputs of 0. In two-dimensional space, such a border is represented by a line. Thus, the AND, OR, NOT, NAND, and NOR gates are linearly separable and can be modeled using a single-layer network. It should be noted that multi-layer neural networks are needed to solve problems associated with nonlinearly separable logic gates. In particular, the XOR-gate is known as the best example of such logic gates; its truth table cannot be represented with any linear combination of input variables since two pairs of input data, (1,0) and (0,1), which provide output equal to one, lay on different sides of any linear line that discriminates two other pairs, (0,0) and (1,1), which provide an output equal to zero. To investigate the functioning of the model of artificial neural networks, it was necessary to analyze the influence of assigning weights, bias values, and threshold values of neuron activity on the logical output. Thus, it was possible to determine what computational mechanism should be reflected in the quiver model and conclude that the artificial neural network model with learned weights does not guarantee logical accuracy.

4.4 Quiver Framework Development

This section represents the first mathematical breakthrough of the project. It lies in expressing the computational architecture of the neural network models of logic gates in terms of quiver representation theory. Every part of the neural network architecture – from the inputs to the outputs, through the weights, biases, and activation functions – was matched by a corresponding term in quiver representation theory. This led to an exact algebraic representation of each logic gate.

General Single-Layer Framework

The single layer quiver representation of a logic gate with n binary inputs is as follows. The vertices are: $Q_0 = \{v_1, v_2, \dots, v_n, b, h, y\}$ where v_i is the binary input signal to the gate, b is the bias, h is the hidden neuron, and y is the logical output. Thus $v_1, v_2, \dots, v_n$ are n input neurons, while h is the hidden neuron, and y is the output neuron.

To understand this particular architectural choice, let us first look at the basic setup of the single layer neural network where the input is directly related to the output without any intermediate nodes:

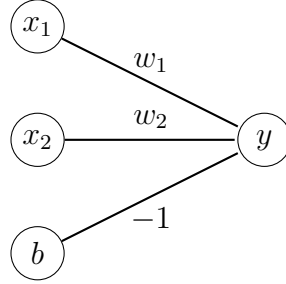


Figure 1: Standard Single-Layer Perceptron Configuration

As compared to this, for achieving the same mathematical result using quiver approach, we have to define h purely as a computational node (or aggregating node):

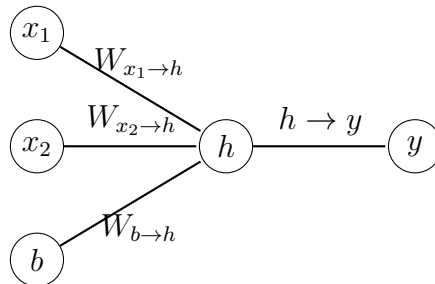


Figure 2: Quiver-Based Framework with Internal Computational Node h

The vertices are connected by arrows α_i 's between input vertex v_i 's and the hidden vertex h , an arrow between b and h , and an arrow between h and y . For each vertex in

the graph, we associate the real numbers $\mathbb{R}$ as their vector spaces.

Each arrow α_i between v_i and h induces a linear map $X(\alpha_i) : \mathbb{R} \rightarrow \mathbb{R}$ given by: $X(\alpha_i)(x_i) = w_i x_i$; where w_i is the weight attached to the arrow α_i . Arrow between h and y induces identity map. Thus, total induced map between the arrows to hidden vertex h , which is the activation sum, is given by:

$$\Phi(x) = \sum_{i=1}^n w_i x_i \quad (3)$$

The output of the gate is thus given by the following equation:

$$y = f(\Phi(x) - b) \quad (4)$$

where b represents the bias and f the step function. If the activation sum equals or surpasses the threshold set by the bias, the gate fires and generates an output of 1. Otherwise, the gate fails to fire, resulting in an output of 0. The operation of each gate depends entirely on the signs and magnitudes of the weights w_i , as well as the bias parameter b . More interestingly, all single-layer gates share the same quiver structure. The diagram of the quiver remains constant regardless of whether we consider an AND, OR, NOT, NAND, or NOR gate. All that differentiates these logic gates is the signs and values associated with the weights and biases.

General Two-Layer Framework

Since XOR and XNOR are not linearly separable, they require a two-layer quiver structure. This follows from the fact that there does not exist any single linear inequality that could separate the inputs producing the output 1, i.e., $(1, 0)$ and $(0, 1)$, from the inputs that produce 0, i.e., $(0, 0)$ and $(1, 1)$.

The sets of vertices in the two-layer quiver structure include:

- Inputs v_1, v_2
- First-layer bias node b_1
- Hidden nodes h_1, h_2
- Second-layer bias node b_2
- Computational node h_3
- Output node y

The set of arrows in the two-layer quiver connect each input to both hidden nodes, b_1 to both hidden nodes, both hidden nodes to the output node, and b_2 to the output node. In the first-layer computation, the hidden activations are calculated as:

$$h_1 = f(x_1 + x_2 - b_1) \quad \text{and} \quad h_2 = f(-x_1 - x_2 + b_1) \quad (5)$$

h_1 makes use of positive weights in order to implement OR functionality, while h_2 makes use of negative weights in order to implement NOR functionality. Thus, they provide us with a new representation in the second layer. The calculation performed by the second layer results in the following output:

$$y = f(h_1 + h_2 - b_2) \quad (6)$$

Both second-layer weights are 1. This leads to adding the results from the hidden neurons. Finally, the bias value b_2 is selected in such a way that the desired result equals 1 only for particular patterns.

4.5 Sensitivity Analysis

The section represents the second important mathematical achievement made by the project. The sensitivity analysis is carried out to determine the effect of bias parameter b on the output logic of the respective gate. The entire range of bias values is determined during the analysis process where the gate operates correctly.

Motivation and Definition

The value of the bias parameter b forms the decision threshold in the step activation function. The increase in b implies a higher value of the activation sum required to trigger the gate, whereas its decrease makes the gate easier to trigger.

Sensitivity analysis is defined as the analysis of stability of the logical operation of a gate with respect to changes in b . The interval of values of the bias parameter that would guarantee deterministic logic operations for any input can be determined using two important values:

- M_0 — maximum value of $\Phi(x)$, where the target output is 0;
- m_1 — minimum value of $\Phi(x)$, where the target output is 1.

The bias interval can be formulated as follows:

$$b \in (M_0, m_1] \quad (7)$$

This means that there exists a certain margin of error for the bias parameter b , so that any value of b in this interval would guarantee correct logic separation by $f(\Phi(x) - b)$.

Single-Layer Bias Condition

For a one-layer quiver gate with induced map $\Phi(x) = \sum w_i x_i$, the following definitions are used:

- $S_1 = \{x \in \{0, 1\}^n : \text{output is 1}\}$
- $S_0 = \{x \in \{0, 1\}^n : \text{output is 0}\}$

In order for the gate to output the desired value for all possible inputs, both the following statements should be true at the same time:

- **Statement 1:** $\Phi(x) - b \geq 0$ for all $x \in S_1$. This means that the gate will always output a value of 1 if its input values imply that its desired output is 1.
- **Statement 2:** $\Phi(x) - b < 0$ for all $x \in S_0$. This means that the gate will not output a value of 1 if its input values imply that its desired output is 0.

After manipulating the above equations, it follows that $b \leq \Phi(x)$ for all $x \in S_1$ and $b > \Phi(x)$ for all $x \in S_0$. As such, the bias should satisfy the **quiver condition**:

$$\max_{x \in S_0} \Phi(x) < b \leq \min_{x \in S_1} \Phi(x) \quad (8)$$

The bias interval is non-empty if and only if:

$$\max_{x \in S_0} \Phi(x) < \min_{x \in S_1} \Phi(x) \quad (9)$$

Two-Layer Bias Condition

In a two-layer gate, the first-layer bias b_1 is obtained by taking the one-layer condition individually on every hidden neuron. Given b_1 takes a specific value from the valid range, we can compute the values of the hidden activations $h_1(x)$ and $h_2(x)$ for every input x .

For the second layer of the network, the induced map is defined as follows:

$$\Psi(h(x)) = u_1 h_1(x) + u_2 h_2(x) \quad (10)$$

where u_1 and u_2 are the weights in the second layer. The second layer bias condition is stated as follows:

$$\max_{x \in S_0} \Psi(h(x)) < b_2 \leq \min_{x \in S_1} \Psi(h(x)) \quad (11)$$

Here, S_1 and S_0 are based on the final logical result desired from the gates.

4.6 Validation

Each gate model was tested by substituting all possible inputs to it and comparing the calculated value of y with the truth table of the gate. In the case of single-layer gates, bias b was selected from the previously found interval $(M_0, m_1]$, while output value was calculated for each input combination as $y = f(\Phi(x) - b)$. In two-layer gates, hidden layer was first activated, second layer map was estimated, and finally, output value was estimated using the second-layer bias b_2 . In all cases, the output values of the quiver gates were always consistent with the Boolean truth tables with perfect accuracy. None of the probabilistic approximation or optimization process was used. The output value of each gate is fully and accurately decided by the following factors:

- Algebraic structure of the quiver Q .
- Weights assigned to the arrows (w_i, u_i) .
- Bias b , which lies within the valid range defined during the sensitivity analysis.

This exact match between the algebraic model of quiver gates and the Boolean truth table proves the fundamental idea of this project, namely that quiver representation theory can be used to describe classical logic gates perfectly accurately.

5 Results & Discussion

The quiver-based framework was applied to all seven fundamental logic gates. For each gate, the quiver structure was constructed, vector spaces and linear maps were assigned, the induced map was computed, and the valid bias interval was determined through sensitivity analysis. The results are presented below.

5.1 Single-Layer Gate Representations

Each single-layer gate is modeled using a quiver $Q = (Q_0, Q_1)$. The general single-layer quiver structure has vertex set:

$$Q_0 = \{v_1, v_2, b, h, y\}$$

and arrow set:

$$Q_1 = \{\alpha_1 : v_1 \rightarrow h, \alpha_2 : v_2 \rightarrow h, \alpha_b : b \rightarrow h, \alpha_h : h \rightarrow y\}$$

Each vertex is assigned the vector space $X(v) = \mathbb{R}$. Each arrow α_i induces a linear map $X(\alpha_i)(x_i) = w_i x_i$, where w_i is the scalar weight assigned to that arrow. The total induced map at the hidden vertex h is:

$$\Phi(x) = \sum_{i=1}^n w_i x_i$$

and the output of the gate is:

$$y = f(\Phi(x) - b), \quad f(z) = \begin{cases} 1 & z \geq 0 \\ 0 & z < 0 \end{cases}$$

AND Gate

Quiver Representation

The AND gate produces an output of 1 only when both inputs are 1. It is represented by the quiver $Q = (Q_0, Q_1)$ with:

$$Q_0 = \{v_1, v_2, b, h, y\}$$

$$Q_1 = \{\alpha_1 : v_1 \rightarrow h, \alpha_2 : v_2 \rightarrow h, \alpha_b : b \rightarrow h, \alpha_h : h \rightarrow y\}$$

Vector space assignment:

$$X(v_1) = X(v_2) = X(b) = X(h) = X(y) = \mathbb{R}$$

Linear maps induced by each arrow:

$$X(\alpha_1)(x_1) = w_1 x_1 = x_1, \quad X(\alpha_2)(x_2) = w_2 x_2 = x_2$$

Both input weights are positive: $w_1 = 1, w_2 = 1$.

Total induced map at vertex h :

$$\Phi(x_1, x_2) = x_1 + x_2$$

Output formula:

$$y = f(x_1 + x_2 - b)$$

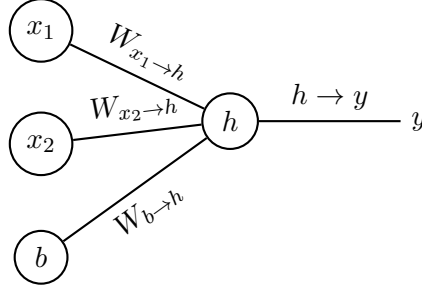


Figure 3: Quiver-based neural network representation of the AND logic gate

Evaluation of Φ over all binary inputs:

x_1	x_2	$\Phi(x_1, x_2)$
0	0	0
1	0	1
0	1	1
1	1	2

Sensitivity Analysis

Set partition from AND truth table:

$$S_1 = \{(1, 1)\}, \quad S_0 = \{(0, 0), (1, 0), (0, 1)\}$$

Bias condition:

$$\max_{x \in S_0} \Phi(x) = 1 < b \leq \min_{x \in S_1} \Phi(x) = 2$$

$$1 < b \leq 2$$

Verification (using $b = 1.5$):

x_1	x_2	$\Phi(x) - b$	$f(\Phi(x) - b)$	Expected
0	0	$0 - 1.5 = -1.5$	0	0
1	0	$1 - 1.5 = -0.5$	0	0
0	1	$1 - 1.5 = -0.5$	0	0
1	1	$2 - 1.5 = 0.5$	1	1

All outputs match the AND truth table exactly.

OR Gate

Quiver Representation

The OR gate produces an output of 1 when at least one input is 1. It uses the same quiver topology as the AND gate:

$$Q_0 = \{v_1, v_2, b, h, y\}, \quad Q_1 = \{\alpha_1 : v_1 \rightarrow h, \alpha_2 : v_2 \rightarrow h, \alpha_b : b \rightarrow h, \alpha_h : h \rightarrow y\}$$

Vector space assignment:

$$X(v_1) = X(v_2) = X(b) = X(h) = X(y) = \mathbb{R}$$

Linear maps induced by each arrow:

$$X(\alpha_1)(x_1) = x_1, \quad X(\alpha_2)(x_2) = x_2$$

Both input weights are positive: $w_1 = 1$, $w_2 = 1$. The logical difference between AND and OR arises entirely from the choice of bias value, not from any change in the quiver structure or weight signs.

Total induced map at vertex h :

$$\Phi(x_1, x_2) = x_1 + x_2$$

Output formula:

$$y = f(x_1 + x_2 - b)$$

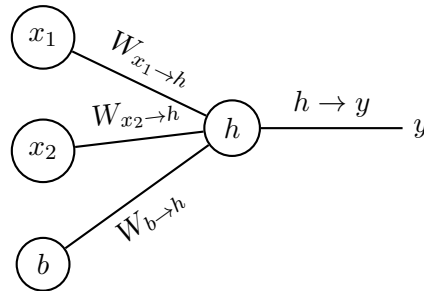


Figure 4: Quiver-based neural network representation of the OR logic gate

Evaluation of Φ over all binary inputs:

x_1	x_2	$\Phi(x_1, x_2)$
0	0	0
1	0	1
0	1	1
1	1	2

Sensitivity Analysis

Set partition from OR truth table:

$$S_1 = \{(1, 0), (0, 1), (1, 1)\}, \quad S_0 = \{(0, 0)\}$$

Bias condition:

$$\max_{x \in S_0} \Phi(x) = 0 \quad < \quad b \quad \leq \quad \min_{x \in S_1} \Phi(x) = 1$$

$$\boxed{0 < b \leq 1}$$

Verification (using $b = 0.5$):

x_1	x_2	$\Phi(x) - b$	$f(\Phi(x) - b)$	Expected
0	0	$0 - 0.5 = -0.5$	0	0
1	0	$1 - 0.5 = 0.5$	1	1
0	1	$1 - 0.5 = 0.5$	1	1
1	1	$2 - 0.5 = 1.5$	1	1

All outputs match the OR truth table exactly.

NOT Gate

Quiver Representation

The NOT gate takes a single input and produces its logical complement. It uses a reduced quiver structure:

$$Q_0 = \{v_1, b, h, y\}, \quad Q_1 = \{\alpha_1 : v_1 \rightarrow h, \alpha_b : b \rightarrow h, \alpha_h : h \rightarrow y\}$$

Vector space assignment:

$$X(v_1) = X(b) = X(h) = X(y) = \mathbb{R}$$

Linear map induced by arrow α_1 :

$$X(\alpha_1)(x_1) = w_1 x_1 = -x_1$$

The weight is negative: $w_1 = -1$. This negative weight ensures that a high input reduces the activation at h , producing logical inversion.

Total induced map at vertex h :

$$\Phi(x_1) = -x_1$$

Output formula:

$$y = f(-x_1 + b)$$

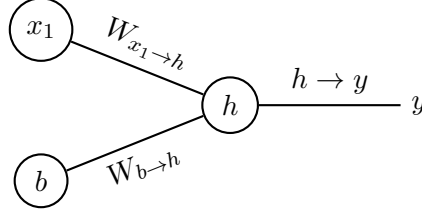


Figure 5: Quiver-based neural network representation of the NOT logic gate

Evaluation of Φ over all binary inputs:

x_1	$\Phi(x_1)$
0	0
1	-1

Sensitivity Analysis

Set partition from NOT truth table:

$$S_1 = \{0\}, \quad S_0 = \{1\}$$

Bias condition:

$$\max_{x \in S_0} \Phi(x) = -1 < b \leq \min_{x \in S_1} \Phi(x) = 0$$

$$-1 < b \leq 0$$

Verification (using $b = -0.5$):

x_1	$\Phi(x) - b$	$f(\Phi(x) - b)$	Expected
0	$0 - (-0.5) = 0.5$	1	1
1	$-1 - (-0.5) = -0.5$	0	0

All outputs match the NOT truth table exactly.

NAND Gate

Quiver Representation

The NAND gate produces an output of 0 only when both inputs are 1, and output 1 in all other cases. It uses the same quiver topology as the AND gate but with reversed weight signs:

$$Q_0 = \{v_1, v_2, b, h, y\}, \quad Q_1 = \{\alpha_1 : v_1 \rightarrow h, \alpha_2 : v_2 \rightarrow h, \alpha_b : b \rightarrow h, \alpha_h : h \rightarrow y\}$$

Vector space assignment:

$$X(v_1) = X(v_2) = X(b) = X(h) = X(y) = \mathbb{R}$$

Linear maps induced by each arrow:

$$X(\alpha_1)(x_1) = -x_1, \quad X(\alpha_2)(x_2) = -x_2$$

Both input weights are negative: $w_1 = -1$, $w_2 = -1$.

Total induced map at vertex h :

$$\Phi(x_1, x_2) = -x_1 - x_2$$

Output formula:

$$y = f(-x_1 - x_2 + b)$$

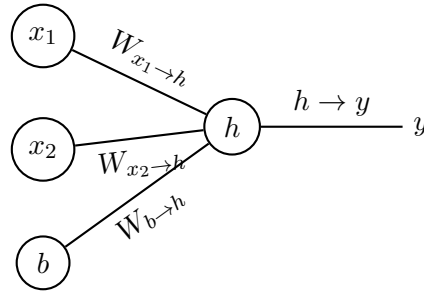


Figure 6: Quiver-based neural network representation of the NAND logic gate

Evaluation of Φ over all binary inputs:

x_1	x_2	$\Phi(x_1, x_2)$
0	0	0
1	0	-1
0	1	-1
1	1	-2

Sensitivity Analysis

Set partition from NAND truth table:

$$S_1 = \{(0, 0), (1, 0), (0, 1)\}, \quad S_0 = \{(1, 1)\}$$

Bias condition:

$$\max_{x \in S_0} \Phi(x) = -2 < b \leq \min_{x \in S_1} \Phi(x) = -1$$

$$\boxed{-2 < b \leq -1}$$

Verification (using $b = -1.5$):

x_1	x_2	$\Phi(x) - b$	$f(\Phi(x) - b)$	Expected
0	0	$0 - (-1.5) = 1.5$	1	1
1	0	$-1 - (-1.5) = 0.5$	1	1
0	1	$-1 - (-1.5) = 0.5$	1	1
1	1	$-2 - (-1.5) = -0.5$	0	0

All outputs match the NAND truth table exactly.

NOR Gate

Quiver Representation

The NOR gate produces an output of 1 only when both inputs are 0. It uses the same quiver topology and negative weights as the NAND gate, differing only in the bias value:

$$Q_0 = \{v_1, v_2, b, h, y\}, \quad Q_1 = \{\alpha_1 : v_1 \rightarrow h, \alpha_2 : v_2 \rightarrow h, \alpha_b : b \rightarrow h, \alpha_h : h \rightarrow y\}$$

Vector space assignment:

$$X(v_1) = X(v_2) = X(b) = X(h) = X(y) = \mathbb{R}$$

Linear maps induced by each arrow:

$$X(\alpha_1)(x_1) = -x_1, \quad X(\alpha_2)(x_2) = -x_2$$

Both weights are negative: $w_1 = -1$, $w_2 = -1$.

Total induced map at vertex h :

$$\Phi(x_1, x_2) = -x_1 - x_2$$

Output formula:

$$y = f(-x_1 - x_2 + b)$$

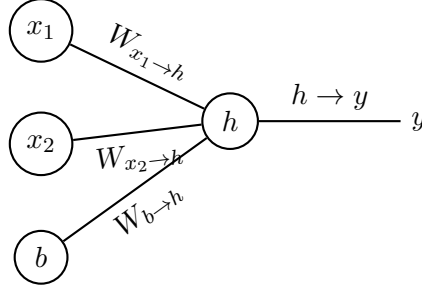


Figure 7: Quiver-based neural network representation of the NOR logic gate

Evaluation of Φ over all binary inputs:

x_1	x_2	$\Phi(x_1, x_2)$
0	0	0
1	0	-1
0	1	-1
1	1	-2

Sensitivity Analysis

Set partition from NOR truth table:

$$S_1 = \{(0, 0)\}, \quad S_0 = \{(1, 0), (0, 1), (1, 1)\}$$

Bias condition:

$$\max_{x \in S_0} \Phi(x) = -1 < b \leq \min_{x \in S_1} \Phi(x) = 0$$

$$\boxed{-1 < b \leq 0}$$

Verification (using $b = -0.5$):

x_1	x_2	$\Phi(x) - b$	$f(\Phi(x) - b)$	Expected
0	0	$0 - (-0.5) = 0.5$	1	1
1	0	$-1 - (-0.5) = -0.5$	0	0
0	1	$-1 - (-0.5) = -0.5$	0	0
1	1	$-2 - (-0.5) = -1.5$	0	0

All outputs match the NOR truth table exactly.

5.2 Two-Layer Gate Representations

The XOR and XNOR gates are not linearly separable. No choice of weights w_1, w_2 and bias b in a single-layer quiver can correctly separate the inputs producing output 1

from those producing output 0 for these gates. A two-layer quiver structure is therefore required.

The general two-layer quiver has vertex set:

$$Q_0 = \{v_1, v_2, b_1, h_1, h_2, b_2, h_3, y\}$$

and arrow set:

$$Q_1 = \left\{ \begin{array}{l} \alpha_1 : v_1 \rightarrow h_1, \alpha_2 : v_2 \rightarrow h_1, \alpha_3 : v_1 \rightarrow h_2, \alpha_4 : v_2 \rightarrow h_2, \alpha_{b_1} : b_1 \rightarrow h_2, \\ \beta_1 : h_1 \rightarrow h_3, \beta_2 : h_2 \rightarrow h_3, \beta_{b_2} : b_2 \rightarrow h_3, \gamma : h_3 \rightarrow y \end{array} \right\}$$

Each vertex is assigned the vector space $\mathbb{R}$. The first layer computes two intermediate activations at h_1 and h_2 , and the second layer combines them at h_3 to produce the final output y .

The first-layer induced maps are:

$$\Phi_1(x_1, x_2) = x_1 + x_2 - b_1 \quad (\text{for hidden neuron } h_1)$$

$$\Phi_2(x_1, x_2) = -x_1 - x_2 + b_1 \quad (\text{for hidden neuron } h_2)$$

The hidden activations are:

$$h_1 = f(x_1 + x_2 - b_1), \quad h_2 = f(-x_1 - x_2 + b_1)$$

The second-layer induced map at h_3 is:

$$\Phi_3(h_1, h_2) = h_1 + h_2 - b_2$$

The final output is:

$$y = f(h_1 + h_2 - b_2)$$

XOR Gate

Quiver Representation

The XOR gate produces an output of 1 when exactly one of the two inputs is 1. It is represented by the two-layer quiver with vertex set:

$$Q_0 = \{v_1, v_2, b_1, h_1, h_2, b_2, h_3, y\}$$

and arrow set:

$$Q_1 = \left\{ \begin{array}{l} \alpha_1 : v_1 \rightarrow h_1, \alpha_2 : v_2 \rightarrow h_1, \alpha_3 : v_1 \rightarrow h_2, \alpha_4 : v_2 \rightarrow h_2, \alpha_{b_1} : b_1 \rightarrow h_2, \\ \beta_1 : h_1 \rightarrow h_3, \beta_2 : h_2 \rightarrow h_3, \beta_{b_2} : b_2 \rightarrow h_3, \gamma : h_3 \rightarrow y \end{array} \right\}$$

Vector space assignment:

$$X(v_1) = X(v_2) = X(h_1) = X(h_2) = X(h_3) = X(y) = \mathbb{R}$$

First-layer linear maps:

The arrows $\alpha_1 : v_1 \rightarrow h_1$ and $\alpha_2 : v_2 \rightarrow h_1$ carry positive weights $+1$, so the induced map at h_1 is:

$$X(\alpha_1)(x_1) = x_1, \quad X(\alpha_2)(x_2) = x_2 \quad \Rightarrow \quad \Phi_1(x_1, x_2) = x_1 + x_2 - b_1$$

The arrows $\alpha_3 : v_1 \rightarrow h_2$ and $\alpha_4 : v_2 \rightarrow h_2$ carry negative weights -1 , so the induced map at h_2 is:

$$X(\alpha_3)(x_1) = -x_1, \quad X(\alpha_4)(x_2) = -x_2 \quad \Rightarrow \quad \Phi_2(x_1, x_2) = -x_1 - x_2 + b_1$$

Hidden activations:

$$h_1 = f(x_1 + x_2 - b_1), \quad h_2 = f(-x_1 - x_2 + b_1)$$

Second-layer linear maps:

The arrows $\beta_1 : h_1 \rightarrow h_3$ and $\beta_2 : h_2 \rightarrow h_3$ carry weights $+1$, so the induced map at h_3 is:

$$X(\beta_1)(h_1) = h_1, \quad X(\beta_2)(h_2) = h_2 \quad \Rightarrow \quad \Phi_3(h_1, h_2) = h_1 + h_2 - b_2$$

Output formula:

$$y = f(h_1 + h_2 - b_2)$$

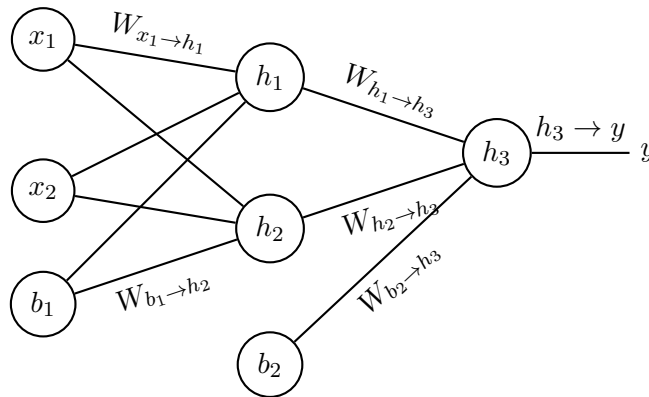


Figure 8: Two-layer quiver-based neural network representation of the XOR logic gate

Evaluation of hidden activations over all binary inputs:

x_1	x_2	$\Phi_1 = x_1 + x_2 - b$	$h_1 = f(\Phi_1)$	$\Phi_2 = -x_1 - x_2 + b$	$h_2 = f(\Phi_2)$
0	0	$-b$	0	b	1
1	0	$1 - b$	1	$b - 1$	0
0	1	$1 - b$	1	$b - 1$	0
1	1	$2 - b$	1	$b - 2$	0

Second-layer evaluation table:

x_1	x_2	h_1	h_2	$\Phi_3 = h_1 + h_2$	Set
0	0	0	1	1	S_0
1	0	1	0	1	S_1
0	1	1	0	1	S_1
1	1	1	0	1	S_0

Sensitivity Analysis

Set partition from XOR truth table:

$$S_1 = \{(1,0), (0,1)\}, \quad S_0 = \{(0,0), (1,1)\}$$

Second-layer bias condition:

$$\max_{x \in S_0} \Phi_3(h(x)) < b_2 \leq \min_{x \in S_1} \Phi_3(h(x))$$

For S_0 : inputs (0,0) and (1,1) both give $\Phi_3 = 1$. $\max_{S_0} \Phi_3 = 0$.

For S_1 : inputs (1,0) and (0,1) both give $\Phi_3 = 1$. $\min_{S_1} \Phi_3 = 1$.

$$0 < b_2 \leq 1$$

Verification (using $b_2 = 0.5$):

x_1	x_2	h_1	h_2	$\Phi_3 - b_2$	$y = f(\Phi_3 - b_2)$
0	0	0	1	$1 - 0.5 = 0.5$	1
1	0	1	0	$1 - 0.5 = 0.5$	1
0	1	1	0	$1 - 0.5 = 0.5$	1
1	1	1	0	$1 - 0.5 = 0.5$	1

XNOR Gate

Quiver Representation

The XNOR gate produces an output of 1 when both inputs are equal. It uses the same two-layer quiver topology as the XOR gate with vertex set:

$$Q_0 = \{v_1, v_2, b_1, h_1, h_2, b_2, h_3, y\}$$

and arrow set:

$$Q_1 = \left\{ \begin{array}{l} \alpha_1 : v_1 \rightarrow h_1, \alpha_2 : v_2 \rightarrow h_1, \alpha_3 : v_1 \rightarrow h_2, \alpha_4 : v_2 \rightarrow h_2, \alpha_{b_1} : b_1 \rightarrow h_2, \\ \beta_1 : h_1 \rightarrow h_3, \beta_2 : h_2 \rightarrow h_3, \beta_{b_2} : b_2 \rightarrow h_3, \gamma : h_3 \rightarrow y \end{array} \right\}$$

Vector space assignment:

$$X(v_1) = X(v_2) = X(h_1) = X(h_2) = X(h_3) = X(y) = \mathbb{R}$$

First-layer linear maps (identical to XOR):

The arrows $\alpha_1 : v_1 \rightarrow h_1$ and $\alpha_2 : v_2 \rightarrow h_1$ carry positive weights +1:

$$X(\alpha_1)(x_1) = x_1, \quad X(\alpha_2)(x_2) = x_2 \quad \Rightarrow \quad \Phi_1(x_1, x_2) = x_1 + x_2 - b_1$$

The arrows $\alpha_3 : v_1 \rightarrow h_2$ and $\alpha_4 : v_2 \rightarrow h_2$ carry negative weights -1:

$$X(\alpha_3)(x_1) = -x_1, \quad X(\alpha_4)(x_2) = -x_2 \quad \Rightarrow \quad \Phi_2(x_1, x_2) = -x_1 - x_2 + b_1$$

Hidden activations:

$$h_1 = f(x_1 + x_2 - b_1), \quad h_2 = f(-x_1 - x_2 + b_1)$$

The XNOR gate shares the same first-layer structure and therefore the same hidden activations as the XOR gate. The difference arises entirely at the second layer at h_3 .

Second-layer linear maps:

The arrows $\beta_1 : h_1 \rightarrow h_3$ and $\beta_2 : h_2 \rightarrow h_3$ carry weights +1:

$$X(\beta_1)(h_1) = h_1, \quad X(\beta_2)(h_2) = h_2 \quad \Rightarrow \quad \Phi_3(h_1, h_2) = h_1 + h_2 - b_2$$

Output formula:

$$y = f(h_1 + h_2 - b_2)$$

Evaluation of hidden activations over all binary inputs:

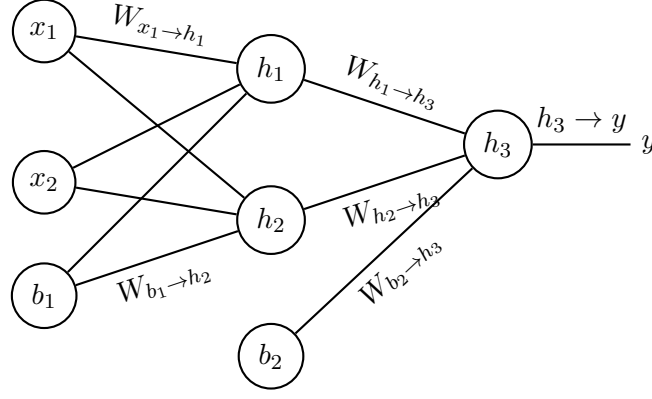


Figure 9: Two-layer quiver-based neural network representation of the XNOR logic gate

x_1	x_2	$\Phi_1 = x_1 + x_2 - b$	$h_1 = f(\Phi_1)$	$\Phi_2 = -x_1 - x_2 + b$	$h_2 = f(\Phi_2)$
0	0	$-b$	0	b	1
1	0	$1 - b$	1	$b - 1$	0
0	1	$1 - b$	1	$b - 1$	0
1	1	$2 - b$	1	$b - 2$	0

Second-layer evaluation table:

x_1	x_2	h_1	h_2	$\Phi_3 = h_1 + h_2$	Set
0	0	0	1	1	S_1
1	0	1	0	1	S_0
0	1	1	0	1	S_0
1	1	1	0	1	S_1

Sensitivity Analysis

Set partition from XNOR truth table:

$$S_1 = \{(0, 0), (1, 1)\}, \quad S_0 = \{(1, 0), (0, 1)\}$$

Second-layer bias condition:

$$\max_{x \in S_0} \Phi_3(h(x)) < b_2 \leq \min_{x \in S_1} \Phi_3(h(x))$$

For S_0 : inputs (1,0) and (0,1) both give $\Phi_3 = 1$. $\max_{S_0} \Phi_3 = 0$.

For S_1 : inputs (0,0) and (1,1) both give $\Phi_3 = 1$. $\min_{S_1} \Phi_3 = 1$.

$$0 < b_2 \leq 1$$

Verification (using $b_2 = 0.5$):

x_1	x_2	h_1	h_2	$\Phi_3 - b_2$	$y = f(\Phi_3 - b_2)$
0	0	0	1	$1 - 0.5 = 0.5$	1
1	0	1	0	$1 - 0.5 = 0.5$	1
0	1	1	0	$1 - 0.5 = 0.5$	1
1	1	1	0	$1 - 0.5 = 0.5$	1

All outputs match the XNOR truth table exactly.

5.3 Summary of Results

The valid bias intervals for all seven logic gates are summarized in the following table.

Gate	Layer	Weights	Induced Map	Bias Range
AND	Single	$w_1 = w_2 = +1$	$\Phi = x_1 + x_2$	$1 < b \leq 2$
OR	Single	$w_1 = w_2 = +1$	$\Phi = x_1 + x_2$	$0 < b \leq 1$
NOT	Single	$w_1 = -1$	$\Phi = -x_1$	$-1 < b \leq 0$
NAND	Single	$w_1 = w_2 = -1$	$\Phi = -x_1 - x_2$	$-2 < b \leq -1$
NOR	Single	$w_1 = w_2 = -1$	$\Phi = -x_1 - x_2$	$-1 < b \leq 0$
XOR	Two	$b_1 = 0.5$	$\Phi_3 = h_1 + h_2$	$0 < b_2 \leq 1$
XNOR	Two	$b_1 = 0.5$	$\Phi_3 = h_1 + h_2$	$0 < b_2 \leq 1$

A non-empty bias interval was obtained for every gate, confirming that exact logical behavior is achievable within the quiver framework for all seven fundamental logic gates. In every case, the width of the valid bias interval is exactly 1, indicating a uniform and symmetric degree of parametric freedom across all gate models. The outputs produced by each quiver representation match the corresponding Boolean truth tables exactly for all binary input combinations, without any approximation at any stage of the computation.

Limitations

Although the proposed framework has been formulated in a very rigorous manner, there are some limitations to the current study. The framework is limited to binary logic, where the input and output signals can be valued either 0 or 1, and neither multi-valued nor fuzzy logic systems have been considered. This study is entirely theoretical and there is no hardware implementation or verification. As a result, problems such as speed, power, and hardware limitations are not considered. The framework has been applied to simple logic gates only, and whether or not the framework will apply to larger digital circuits and combinational circuits has not been investigated. This study has employed only step activation functions, and the impact of different types of activation functions on the accuracy and generality of the framework has not been explored. The framework has

been applied only to combinational logic, and sequential logic components such as flip flops and registers are not included in the study. Finally, the computational complexity of constructing and analyzing large quivers has not been considered.

6 Conclusion

In this work, an algebraic framework is presented for modeling classical logic gates using the quiver representation theory. Through the construction of vector space representations for inputs, hidden layers, and output layers of logical circuits, logical operations can be precisely described using a structured algebraic notation. This model offers a uniform viewpoint on how logical decisions can be made based on algebraic computation rather than probabilistic learning or symbolic reasoning.

All seven basic logic gates AND, OR, NOT, NAND, NOR, XOR, and XNOR gates can be efficiently modeled using the quiver representation theory. All single-layer logic gates can be expressed in terms of a quiver model consisting of input vertices, bias vertex, and hidden vertex connected by arrows representing scalar linear mappings. For two-layer logic gates XOR and XNOR, which are nonlinearly separable, they can be expressed in terms of a two-layer quiver consisting of hidden layer computing the intermediate representations making them linearly separable in the second layer. In all cases, the quiver models accurately produce the same results as their Boolean truth tables.

The sensitivity analysis conducted in the current project studied how the value of the bias parameter affects the logical value output of each gate and found the valid interval of bias values for each gate. All gates have non-empty intervals of bias, demonstrating that it is possible to achieve exact logical values for all seven fundamental logic gates through their representations as a quiver. The fact that the width of the interval of bias values is always equal to one represents an intrinsic algebraic property of the representation of gates and quantifies their parametric flexibility.

This theoretical model guarantees not only correctness but also allows for further insightful analysis of the behavior of the logical system. The model conducts sensitivity analysis on the effect of variations in the bias parameter on the output of the logical system, and identifies the permissible values of the bias for each logic gate. This approach provides assurance of accuracy and reliability of the logical system. Overall, this work shows the practical applications of quivers in contemporary mathematics and opens new research opportunities at the intersection of mathematics and computer science.

7 Further scope of Research

This project framework offers a solid starting point for some interesting directions in future research. Firstly, the construction can be scaled up from single gates to complicated combinational circuits, like adders and decoders, using combinations of quivers whose last vertex becomes the first vertex in the next gate. Expanding on sequential elements such as flip-flops and finite state machines involves including an extra dimension in the quiver structure corresponding to time. Moreover, the quiver framework lends itself naturally to a method for algebraically verifying circuit design based on the concept of morphism and isomorphism of representations. Another interesting direction would involve extending the proposed framework to other logic structures. Multi-valued logic can be achieved by using a vector space of inputs that is not binary, while fuzzy logic requires using truth values in the continuous range $[0, 1]$. Also, it is important to consider what happens when the quiver construction utilizes Sigmoid or ReLU activations instead of the standard step function. Lastly, due to the similarities in structure between linear maps for quiver representations and unitary operators, it is possible to adapt this method to model and verify quantum logic gates, which provides an innovative aspect to quantum circuit design.

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